Vidya Pratishthan's Kamalnayan Bajaj Institute of Engineering and Technology, Baramati

A Project Report On

"COLLEGE PLACEMENT & INTERNSHIP SYSTEM"

Submitted in the partial fulfillment of the requirements

Second Year of Engineering in Information Technology

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2025- 2026

VPKBIET, BARAMATI

Certificate

This is to certify that the Project Report on

"COLLEGE PLACEMENT & INTERNSHIP SYSTEM"

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in the partial fulfilment of the requirement for the successful completion of Community Engineering Project in third year B. Tech (Information Technology) at Vidya Pratishthan's Kamalnayan Bajaj Institute of Engineering and Technology, Baramati, under the Savitribai Phule Pune University, Pune. This work is done during year 2025-2026 Semester-II, under our guidance.

Ms. Y.N. Sakhare	Dr. S.A.Takale	Dr. S.B.Lande
Project Guide	Head of Dept.	Principal

Examiner 1:

Examiner 2:

Acknowledgement

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Abstract

The **College Placement and Internship System** is a web-based application designed to simplify and automate the placement and internship process within educational institutions. It provides an integrated platform for students, companies, and administrators to interact efficiently and manage recruitment activities in a systematic manner.

The primary objective of this system is to overcome the limitations of traditional manual placement processes, such as excessive paperwork, lack of transparency, and time-consuming procedures. Through this platform, students can register, create their profiles, upload resumes, and apply for job and internship opportunities. Companies can register, post job openings, and review applications submitted by students. The administrator is responsible for managing and monitoring the overall system to ensure its smooth functioning.

The system is developed using modern web technologies, including HTML, CSS, and JavaScript for the frontend, PHP for backend processing, and SQLite for database management. It incorporates secure login mechanisms, role-based access control, and efficient data handling techniques.

This project improves communication between students and recruiters, enhances the efficiency of placement activities, and provides a centralized platform for managing recruitment data. Overall, the system offers a reliable, user-friendly, and scalable solution for effective college placement and internship management.

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1. Introduction

1.1 Introduction

The College Placement and Internship System is a web-based application developed to streamline and automate the placement and internship processes in educational institutions. In the current competitive environment, managing placement activities manually is challenging, time-consuming, and inefficient. This system provides a centralized digital platform that connects students, companies, and administrators in an organized and effective manner.

The application enables students to create profiles, upload resumes, and apply for job and internship opportunities. Companies can post job openings, review applications, and select candidates based on their requirements. The administrator is responsible for managing and monitoring all activities within the system, ensuring smooth operation and maintaining data integrity.

By utilizing modern web technologies such as HTML, CSS, JavaScript, PHP, and database systems like SQLite, the system ensures secure data handling, easy accessibility, and improved communication among all stakeholders. Overall, the system enhances efficiency, transparency, and reliability in the placement process.

1.2 Objectives

The main objectives of the College Placement and Internship System are as follows:

- To automate the placement and internship process
- To reduce manual work and paperwork
- To provide a centralized platform for students, companies, and administrators
- To enable students to easily apply for jobs and internships
- To allow companies to efficiently manage job postings and applications
- To maintain accurate and secure records of students and recruitment data

- To improve communication between students and recruiters

1.3 Motivation

The motivation behind developing the College Placement and Internship System arises from the challenges associated with traditional placement processes. In many institutions, placement activities are still handled manually, which leads to inefficiencies, delays, and errors in data management.

Students often face difficulties in accessing timely information about job opportunities, while companies encounter challenges in managing a large number of applications. Additionally, administrators are burdened with complex record-keeping and coordination tasks.

With the rapid advancement of technology and the growing need for digital solutions, there is a demand for a system that can simplify and automate these processes. This project aims to address these challenges by providing a user-friendly and efficient platform that enhances productivity, reduces workload, and ensures transparency in placement activities.

1.4 Problem Statement

The existing placement and internship management systems in many colleges rely heavily on manual processes, which are inefficient and prone to errors. These systems often lack proper organization, transparency, and real-time communication, making it difficult for students, companies, and administrators to interact effectively.

Students may miss opportunities due to a lack of timely updates, companies may struggle to identify suitable candidates, and administrators may find it difficult to manage large volumes of data. Furthermore, manual record-keeping increases the risk of data loss and inconsistency.

Therefore, there is a need for a web-based system that can efficiently automate and manage placement and internship activities. The proposed system aims to provide a centralized, secure, and user-friendly platform to overcome these challenges and improve the overall placement process.

2. Methodology

2.1. Method Summarization

The methodology followed for the development of the College Placement and Internship System is based on a systematic and structured approach to ensure efficiency and accuracy. The development process follows the Software Development Life Cycle (SDLC) model, specifically the Waterfall Model, where each phase is completed before moving to the next.

The key phases involved are:

- **Requirement Analysis:**
- Understanding the needs of students, companies, and administrators. Identifying system functionalities such as registration, job posting, application tracking, and resume management.
- **System Design:**
- Designing the overall structure of the system including database design, user interface layouts, and system architecture.
- **Implementation (Coding):**
- Developing the frontend using HTML, CSS, and JavaScript, and backend using PHP. Database operations are handled using SQLite/MySQL.
- **Testing:**
  Verifying the system for errors, bugs, and performance issues. Ensuring all modules work correctly.
- **Deployment:**
  Hosting the system on a local server (XAMPP/WAMP) or live server for user access.
- **Maintenance:**
  Updating the system based on user feedback and fixing any issues that arise.

This structured methodology ensures that the system is reliable, scalable, and easy to maintain.

2.2. Architecture

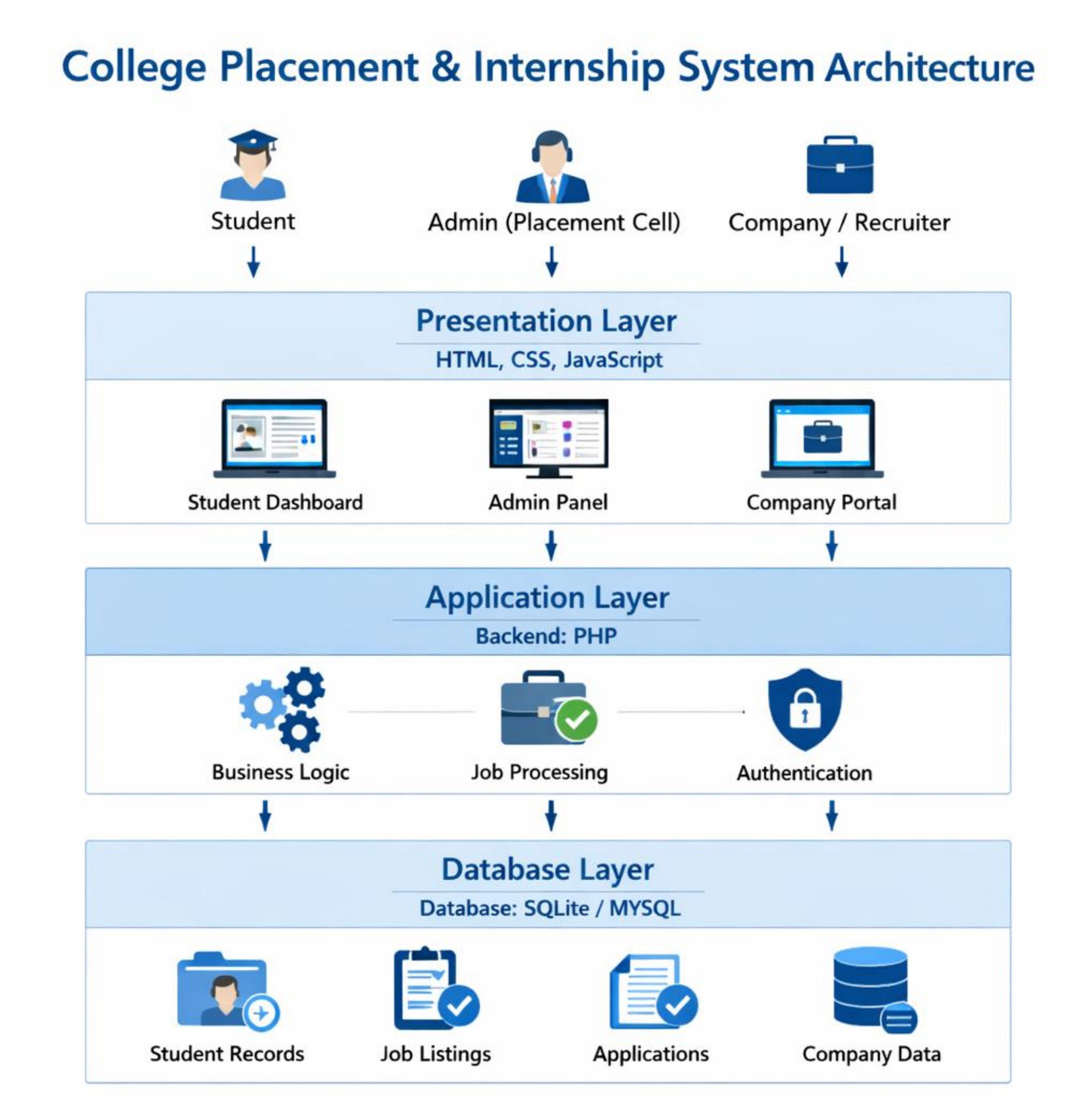


Fig. Architecture Diagram

2.3 Architecture Information

The given diagram represents the Three-Tier Architecture of the College Placement and Internship System. It shows how different users interact with the system and how data flows through various layers.

1. Users of the System

The system is accessed by three main types of users:

- **Student:**

  Students can register, log in, view job/internship opportunities, upload resumes, and apply for jobs.

- **Admin (Placement Cell):**

  The admin manages the entire system, including student records, company registrations, and job postings.

- **Company / Recruiter:**

  Companies can post job openings, view applications, and select candidates.

These users interact with the system through the Presentation Layer.

2. Presentation Layer (Frontend)

This is the **top layer** of the system, developed using:

- HTML
- CSS
- JavaScript

It provides the user interface for all users. It includes:

- **Student Dashboard:** For students to manage profiles and applications
- **Admin Panel:** For system management and monitoring

- **Company Portal:** For recruiters to post jobs and review candidates

This layer collects user input and sends it to the application layer for processing.

3. Application Layer (Backend)

This is the middle layer, implemented using PHP. It acts as a bridge between the frontend and the database.

Main functionalities include:

- **Business Logic:** Controls system operations and workflows
- **Job Processing:** Handles job postings, applications, and status updates
- **Authentication:** Manages login, session control, and user verification

This layer processes user requests, performs validations, and communicates with the database.

4. Database Layer (Data Storage)

This is the bottom layer, which uses SQLite to store all system data.

It manages:

- **Student Records:** Personal details, academic data, resumes
- **Job Listings:** Information about job and internship opportunities
- **Applications:** Records of student applications and their status
- **Company Data:** Details of registered companies

This layer ensures data is stored securely and can be retrieved efficiently.

5. Working Flow of the System

1. The user interacts with the system through the frontend (Presentation Layer)

2. The request is sent to the backend (Application Layer)
3. The backend processes the request using business logic
4. Data is fetched from or stored in the database (Database Layer)
5. The response is sent back to the user interface

6. Key Benefits of This Architecture

- Separation of concerns (UI, logic, data)
- Easy maintenance and scalability
- Improved security and performance
- Flexibility to upgrade individual layers

3. Algorithms & Design

This section explains the algorithms and design approach used in developing the College Placement and Internship System. The system is designed using modular programming and follows a structured approach to ensure efficiency, reliability, and scalability.

3.1 System Design Approach

The system follows a modular design approach, where the entire application is divided into smaller functional modules such as:

- User Authentication
- Student Management
- Company Management
- Job Posting
- Application Processing

Each module is developed and tested independently, making the system easier to maintain and upgrade.

3.2 Algorithms Used

1. User Registration Module

Steps:

1. Start
2. User enters registration details (name, email, password, etc.)
3. Validate input data
4. Check if email already exists in the database
5. If exists → Show error message
6. Else → Store user data in database

7. Display success message
8. End

2. User Login Module (Authentication)

Steps:

1. Start
2. User enters email and password
3. Validate input fields
4. Retrieve user data from database
5. Compare entered password with stored password
6. If matched → Allow login and create session
7. Else → Display error message
8. End

3. Job Posting Module (Company)

Steps:

1. Start
2. Company logs into the system
3. Enter job details (title, description, eligibility, etc.)
4. Validate input data
5. Store job information in database
6. Display confirmation message
7. End

4. Job Application Module (Student)

Steps:

1. Start
2. Student logs into the system
3. Browse available jobs
4. Select a job and click “Apply”
5. Check if already applied
6. If yes → Show message
7. Else → Store application in database
8. Update application status as “Applied”
9. End

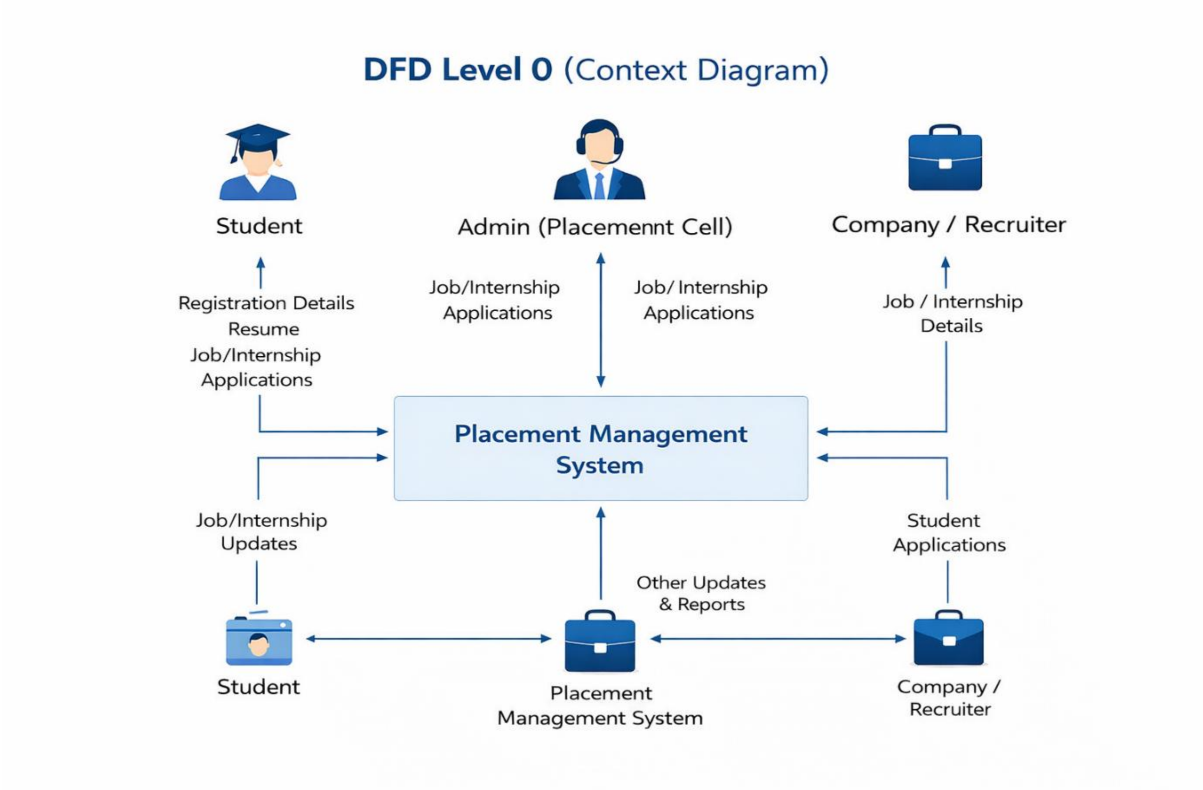
5. Application Selection Module

Steps:

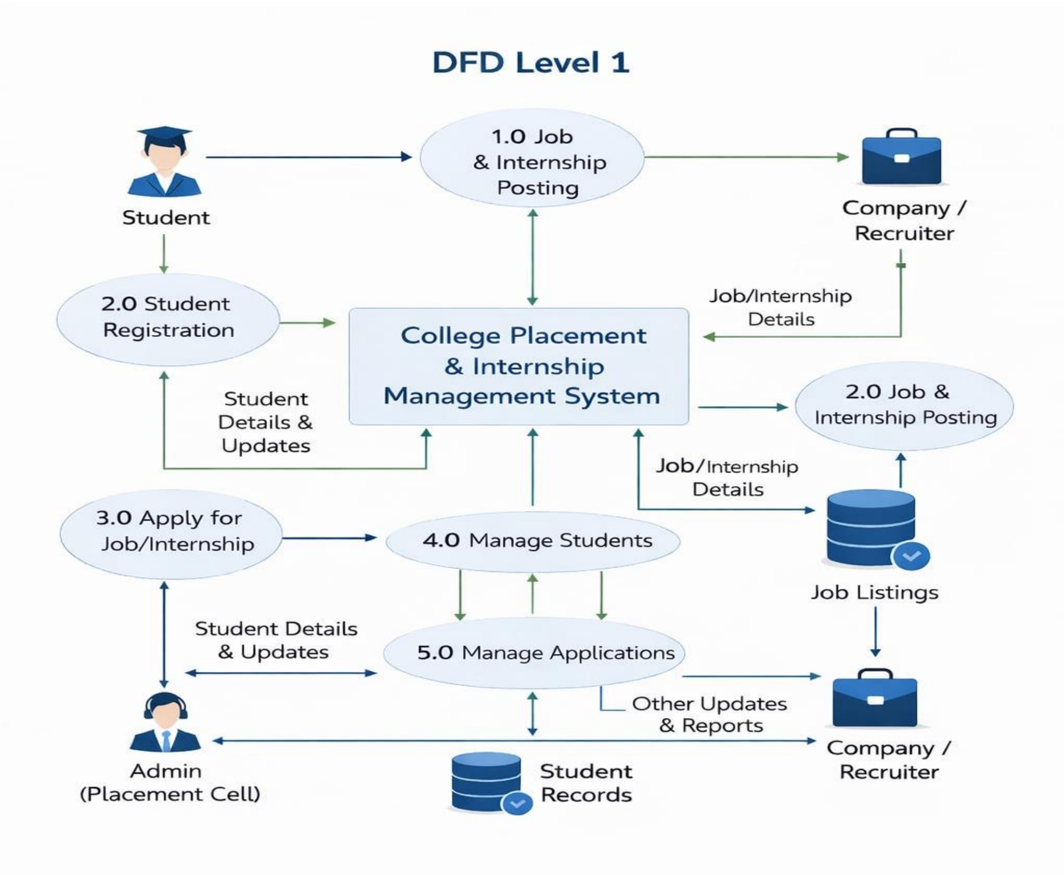
1. Start
2. Company views list of applicants
3. Filter candidates based on criteria
4. Select suitable candidates
5. Update application status (Selected/Rejected)
6. Notify students
7. End

3.3 Data Flow Design

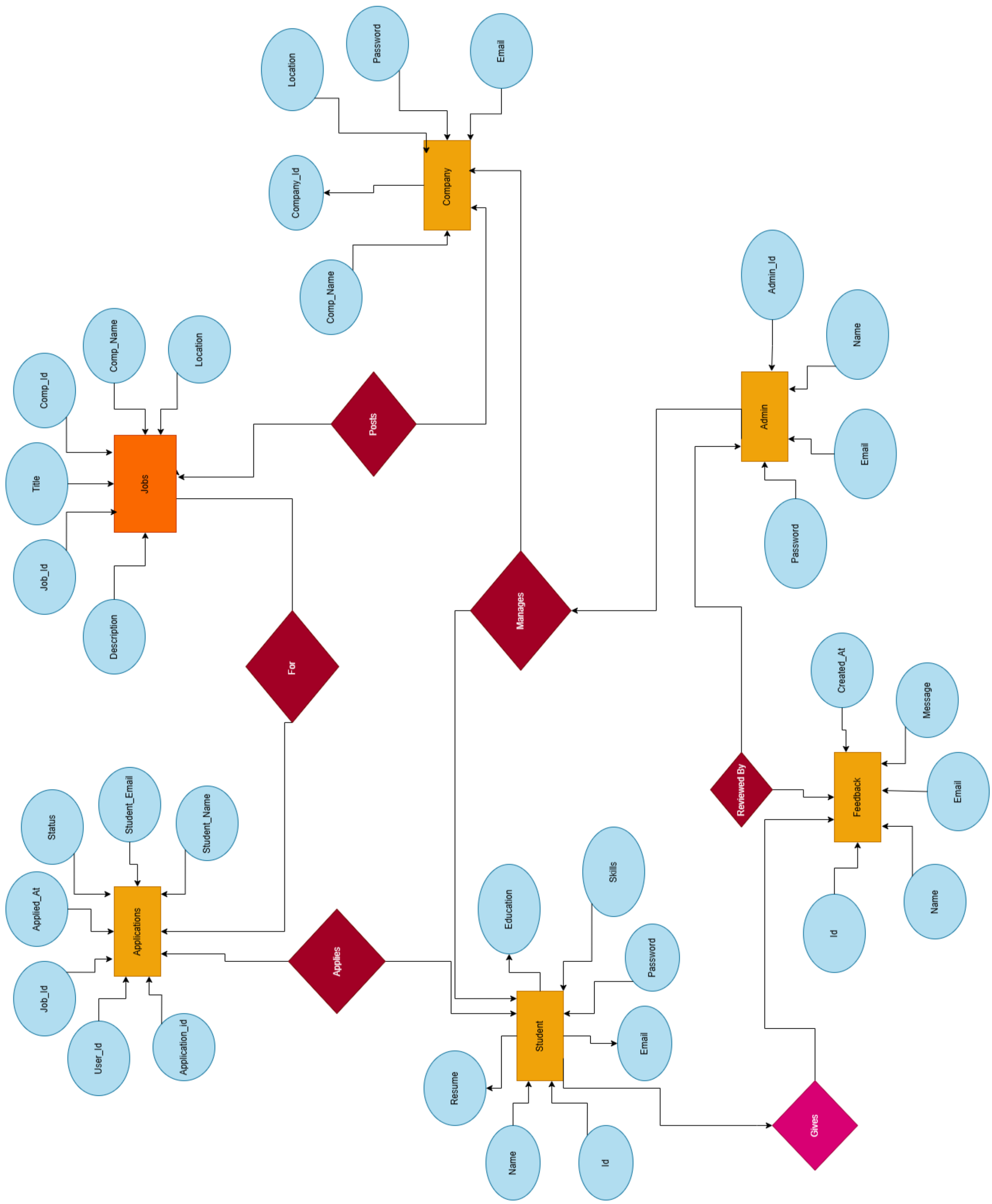
1) DFD level 0



2) DFD level 1

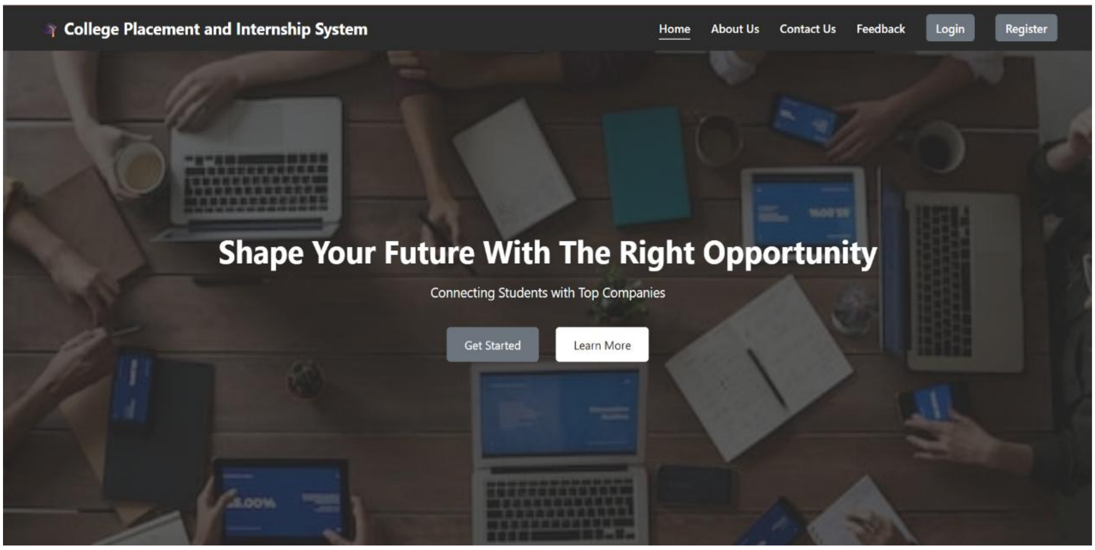


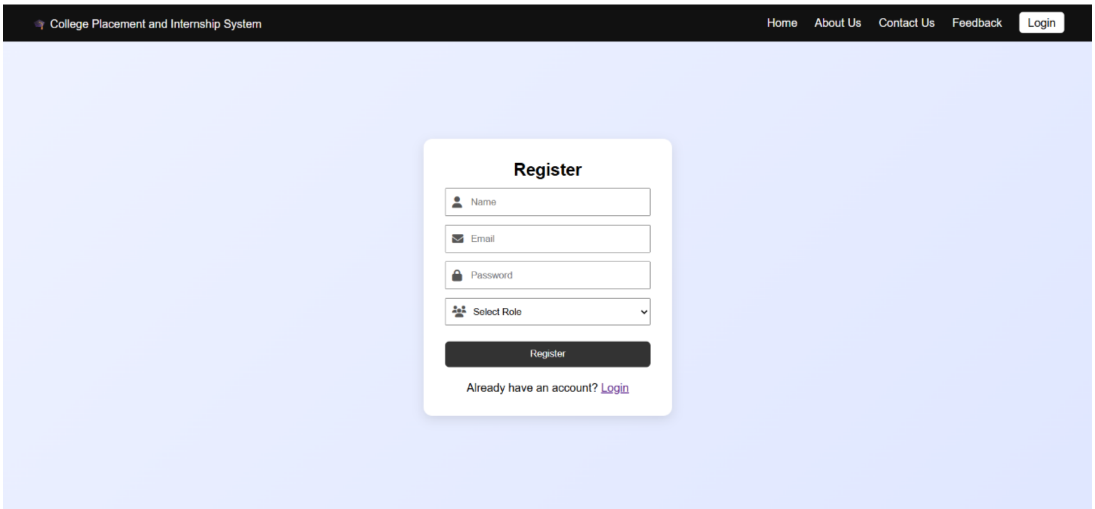
3.4 ER Diagram

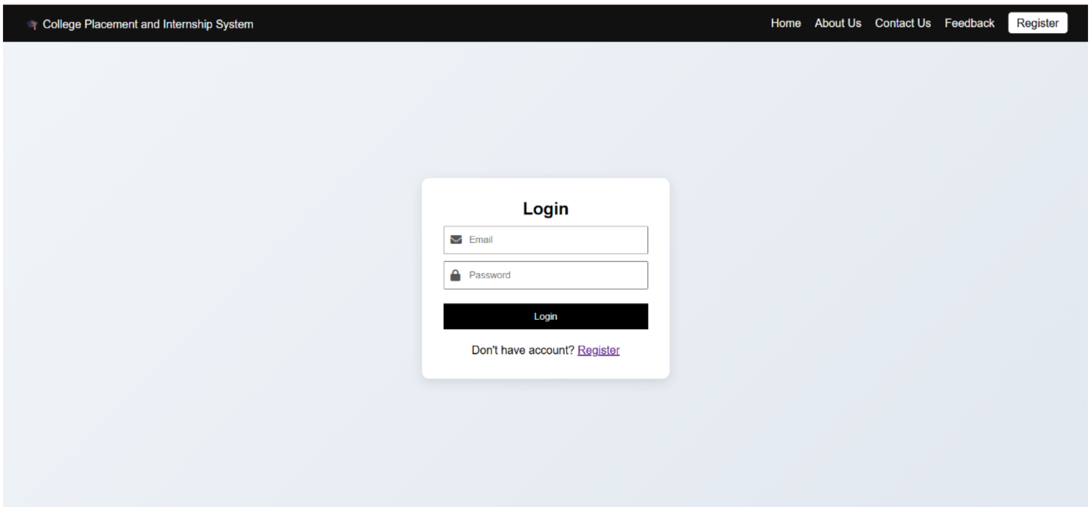


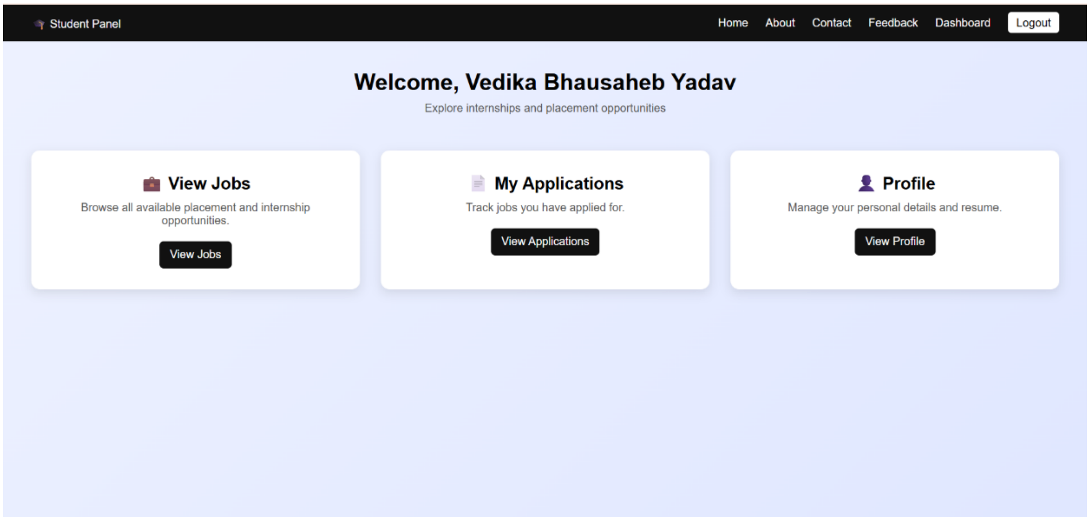
4. Results and Discussion

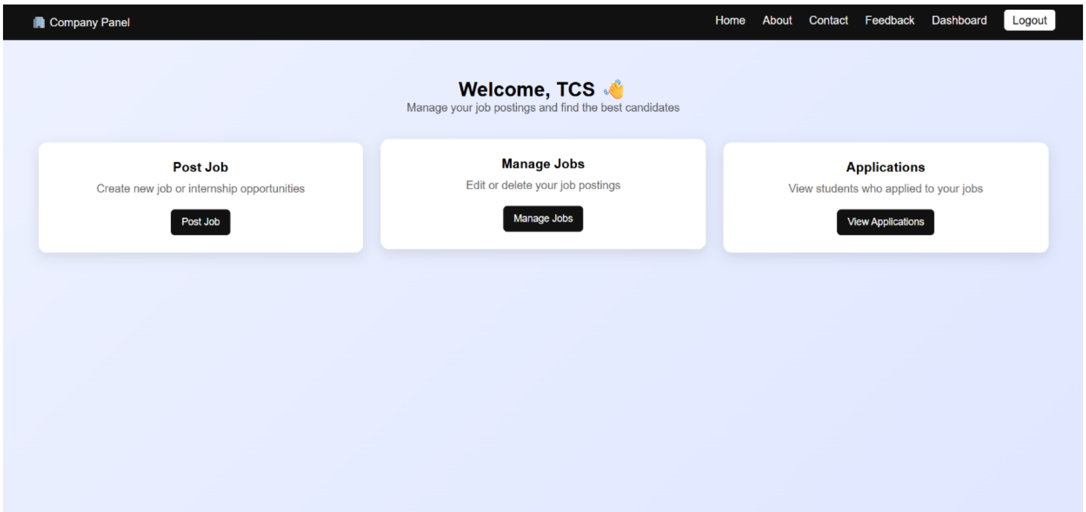
4.1. Results

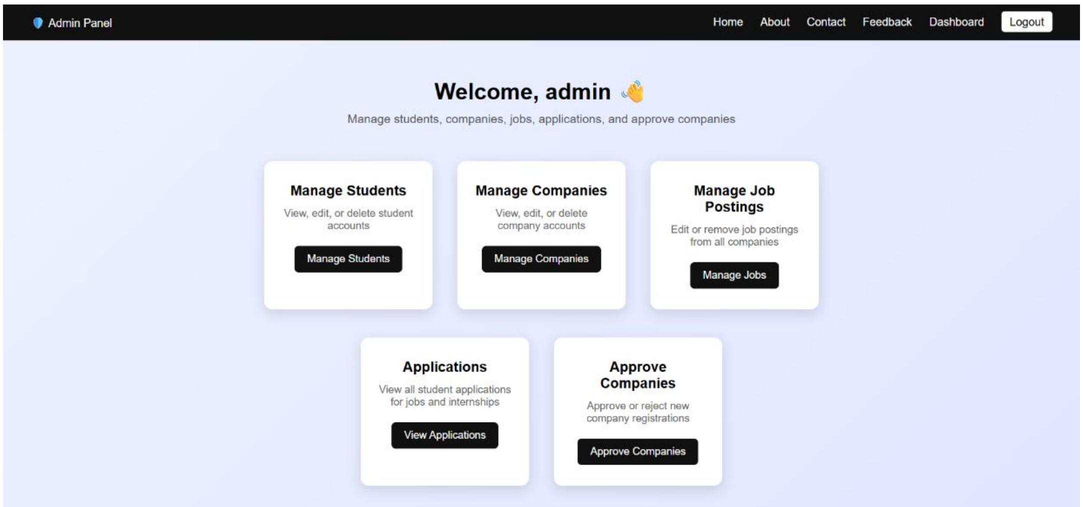







Student Panel
Home
About
Contact
Feedback
Dashboard
Logout
Welcome, Vedika Bhausaheb Yadav
Explore internships and placement opportunities
View Jobs
Browse all available placement and internship opportunities.
View Jobs
My Applications
Track jobs you have applied for.
View Applications
Profile
Manage your personal details and resume.
View Profile


Company Panel
Home
About
Contact
Feedback
Dashboard
Logout
Welcome, TCS
Manage your job postings and find the best candidates
Post Job
Create new job or internship opportunities
Post Job
Manage Jobs
Edit or delete your job postings
Manage Jobs
Applications
View students who applied to your jobs
View Applications


Admin Panel
Home
About
Contact
Feedback
Dashboard
Logout
Welcome, admin
Manage students, companies, jobs, applications, and approve companies
Manage Students
View, edit, or delete student accounts
Manage Students
Manage Companies
View, edit, or delete company accounts
Manage Companies
Manage Job Postings
Edit or remove job postings from all companies
Manage Jobs
Applications
View all student applications for jobs and internships
View Applications
Approve Companies
Approve or reject new company registrations
Approve Companies

4.2. Discussion

The developed system improves the traditional placement process by automating important activities. It reduces manual work, minimizes errors, and enables faster communication between students and recruiters.

Compared to manual systems:

- It provides real-time updates
- Ensures data accuracy and consistency
- Improves transparency in the selection process
- Makes the system accessible anytime and anywhere

However, some challenges were observed:

- It requires a stable internet connection
- Proper data security measures are needed
- Performance may decrease with large amounts of data

4.3. Performance Analysis

- The system performs well for small to medium-sized data
- Response time is fast for user requests
- Database operations are efficient for basic queries

Conclusion & Future Scope

Conclusion

The College Placement and Internship System has been successfully developed to simplify and automate the placement process. It provides a centralized platform for students, companies, and administrators, reducing manual work and improving efficiency. The system offers features like registration, job posting, and application management, ensuring better organization and communication. Overall, it is a reliable and user-friendly solution, with scope for future improvements such as advanced features and mobile support.

Future Scope

The College Placement and Internship System can be further enhanced by adding advanced features to improve its functionality and user experience. Future improvements may include automated notifications through email or SMS to keep users updated about job opportunities and application status. The system can also be upgraded with AI-based recommendations to suggest suitable jobs based on student profiles. Additionally, developing a mobile application will make the system more accessible and convenient. Enhancing security measures, improving performance for handling large data, and integrating with external platforms can further increase the efficiency and scalability of the system.

References

1. Thomas H. Cormen, Charles E. Leiserson, Ronald L. Rivest, and Clifford Stein, "Introduction to Algorithms," MIT Press, 2009.

2. Rasmus Lerdorf and Kevin Tatroe, "Programming PHP," O'Reilly Media, 2013.

3. Robin Nixon, "Learning PHP, MySQL & JavaScript," O'Reilly Media, 2018.

4. "PHP Tutorial," W3Schools, Available: https://www.w3schools.com/php/, Accessed on: March 2026.

5. "HTML, CSS, and JavaScript Documentation," MDN Web Docs, Available: https://developer.mozilla.org/, Accessed on: March 2026.

6. "MySQL Documentation," Oracle Corporation, Available: https://dev.mysql.com/doc/, Accessed on: March 2026.

7. "SQLite Documentation," SQLite.org, Available: https://www.sqlite.org/docs.html, Accessed on: March 2026.

8. "Web-Based Placement Management System," International Journal of Computer Applications (IJCA), 2020.

9. Ian Sommerville, Software Engineering, 10th Edition, Pearson Education, 2015.

10. Roger S. Pressman, Software Engineering: A Practitioner's Approach, 7th Edition, McGraw-Hill Education, 2010.